00 — Squadron Colby: Team and Roles

Mission: 04 — Direct Preference Optimization (DPO), Verbosity Regularization & Steering Vector Control
Course: Conversational AI: History, Applications, Future · WiSe 2026/27 · BTU Cottbus-Senftenberg
Repository: https://github.com/xxrafidxx/Squadron-Colby---Conversational-AI
Waypoint: Betelgeuse & Orion Molecular Cloud Complex (`SEC-04-ORI`, 642 ly from Sol)
AURA Subsystem Repair: Subsystem 04: *Plasma Ignition & Ethical Steering Core* (+7.1% → 28.4% total integrity)

Team

Name	Squadron station	Mission 04 role	Deliverable
Rafid Mohammad Ahsan	Core & Lead Architect	Part 1 · Systems Architect (Lead Engineer)	`01_DPO_Loss_Core.py`, `ps04_solution.ipynb`
Rafid Mohammad Ahsan	Protocol & Interface Specialist	Part 2 · Communications & Protocol Officer	`02_Implicit_Reward_Evaluator.py`
Tahjib Al Sakin	Data & Memory Engineer	Part 3 · Data & Payload Specialist	`03_Length_Bias_and_SimPO.py`
Kunal Choudhary	Security & Ethical Navigator	Part 4 · Security & Ethics Sentinel	`04_Activation_Steering_and_Audit.py`, `04_Alignment_Tax_and_Steering_Audit.md`
Mubina Rahaman Jerin	AI Security Officer	EU AI Act compliance	`06_AI_Act_Compliance.md`

Who Did What

Name	Work done
Rafid Mohammad Ahsan	*Architect:* built `DPOLoss` and `get_batch_logps` in pure PyTorch (gather + `-100` masking, `logsigmoid`); built the AURA Lab testbed (`aura_lab.py`) and ran real DPO: start loss = ln 2, only policy weights get gradients (41/41 vs 0/41), held-out honest correction 48.5 % → 78.8 %; showed that without "agree" pairs DPO learns *always say no*. Assembled the notebook. *Protocol:* implicit rewards of the trained policy (pairwise accuracy 81.8 %), reliability diagram against a simulated rater panel, ECE 0.131 → 0.121 with temperature scaling.
Tahjib Al Sakin	Set up the Mission 04 folder and the `SimPOLoss` module; trained DPO, SimPO and SimPO+SFT on length-confounded data and measured length bias (Pearson r +0.58 / +0.67 / +0.47); measured memory: −50 % weights, −20 % training state without the reference model.
Kunal Choudhary	Extracted the honest-minus-sycophantic vector from the model's activations, injected it with `ActivationSteeringHook`, scanned all layers and swept α from −4 to 20; measured window α ∈ [7, 9] (honest 84.8 %, arithmetic 100 %); wrote `04_Alignment_Tax_and_Steering_Audit.md`.
Mubina Rahaman Jerin	Reviewed the mission under the EU AI Act (Art. 15, 50, 55): flagged the *always say no* shortcut, length growth after DPO and the need to clamp α; wrote `06_AI_Act_Compliance.md`.

Moodle submission (3 files): `00_Team_Roles.pdf` · `Mission04_Role_Deliverables_01-04.pdf` (Parts 1–4, one section per role) · `06_AI_Act_Compliance.pdf`. Each role document is also a separate PDF in the repository (`mission-04/pdf/`).